Faculty of Science	**Linear Algebra and Calculus**
Department of Mathematics	Final Exam 2025
VU	16-12-2025, 08:30–11:15

Name: ______________________________

Student ID: ______________________________ **Grade:** ______

Full Exam or only Linear Algebra? ______________
WARNING: Only the questions of the exam chosen above will be graded.

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	2½	2	2½	2	5	4	6	5	4	3	36
Score:											

Calculus Grade = #points of questions (1 - 4) + 1 = ____

$$\text{Linear Algebra Grade} = \frac{\#\text{points of questions (5 - 10)}}{3} + 1 = \_\_\_\_$$

If Full Exam chosen, Final Grade = Calculus Grade * 0.3 + Linear Algebra Grade * 0.7

Read the instructions carefully!

- The use of a calculator, book or notes is **not** allowed.
- Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
- You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
- Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.
- If you feel like you will be short in time, prioritize questions you know or the ones with higher score count.

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1. **(2 ½ pts.) Calculus:** A particle moves along a line with position function

$$s(t) = t^2 e^{-t}, \quad t \geq 0.$$

Find the times when the particle is at rest and determine whether it is speeding up or slowing down at $t = 1$.
Hint: The velocity function of the particle is given by the derivative of the position function, and the acceleration function is given by the derivative of the velocity function.

2. **Calculus:** Let

$$f(x, y, z) = x^2 y + yz^3 - 4xz.$$

(a) **(1 pts.)** Compute the gradient $\nabla f(x, y, z)$.

(b) **(1 pts.)** Find the directional derivative of f at the point $P = (1, 2, -1)$ in the direction of the vector $\mathbf{v} = (2, -2, 1)$.

3. **(2½ pts.) Calculus:** Consider the following piecewise function:

$$f(x) = \begin{cases} \frac{kx^2 - 9k}{x-3} & x < 3 \\ 2x^2 - 16 & x \geq 3 \end{cases}$$

For which values of the parameter k is f continuous?

4. **(2 pts.) Calculus:** Show that there exist solutions to the equation $x^3 + \sin x = 3x$ in the interval $[-2, -1]$.

5. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by

$$T(x, y, z) = (x + y,\ 2x - z,\ y + z).$$

(a) **(1 pts.)** Find the standard matrix of T.

(b) **(3 pts.)** Compute a basis for the kernel of T.

(c) **(1 pts.)** Without doing any computations, find a basis for the image (or range) of T.

6. Let $A = \dfrac{1}{8}\begin{bmatrix} 4 & -2 & -2 \\ -2 & 5 & 1 \\ -2 & 1 & 5 \end{bmatrix}$.

(a) **(3 pts.)** Compute A^{-1} if it exists.

(b) **(1 pts.)** Assume $B \neq 0$ is a square matrix such that $B^2 = 0$. Show that B is not invertible.
Hint: Assume B is invertible and reach a contradiction.

7. Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \, : \, x + y + z = 0, \, 2x - y = 0\}.$$

(a) **(3 pts.)** Show that W is a vector subspace of $\mathbb{R}^3$.

(b) **(2 pts.)** Find a basis of W.

(c) **(1 pts.)** Which is the dimension of W?

8. Let $A = \begin{bmatrix} 4 & 1 & -2 \\ 0 & 3 & 0 \\ 2 & 1 & 0 \end{bmatrix}$.

(a) **(1 pts.)** Find the eigenvalues of A.

(b) **(3 pts.)** For each eigenvalue, compute a basis of its eigenspace.

(c) **(1 pts.)** Determine whether A is diagonalizable. If so, give the diagonalization, i.e., an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. If not, explain why A is not diagonalizable.

9. Let

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

(a) **(3 pts.)** Find an orthogonal basis for the space spanned by v_1, v_2, v_3.

(b) **(1 pts.)** Find the orthogonal projection of $y = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ onto the space spanned by v_1, v_2, v_3.

10. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

    (a) **(1 pts.)** If A is an invertible matrix, then $\det A^{-1} = \frac{1}{\det A}$.

    (b) **(1 pts.)** Let A and B be two square matrices. Then in general $(A+B)^2 = A^2 + 2AB + B^2$.

    (c) **(1 pts.)** The set $\{1, t^2, t^4\}$ is a basis for $\mathbb{P}_4$.